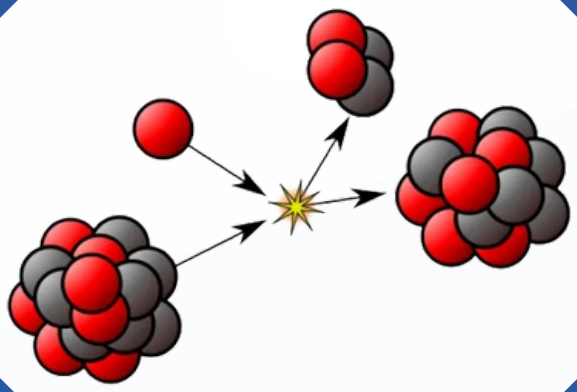
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

path to success

JEE (MAIN+ADVANCED)

Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Modern Physics-1

English Medium

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MODERN PHYSICS-1

KEY CONCEPT

PHOTO ELECTRIC EFFECT

PHOTOELECTRIC EFFECT

It was discovered by Hertz in 1887. He found that when the negative plate of an electric discharge tube was illuminated with ultraviolet light, the electric discharge took place more readily. Further experiments carried out by Hallwachs confirmed that certain negatively charged particles are emitted, when a Zn plate is illuminated with ultraviolet light. These particles were identified as electrons.

The phenomenon of emission of electrons from the surface of certain substances, when suitable radiations of certain frequency or wavelength are incident upon it is called photoelectric effect.

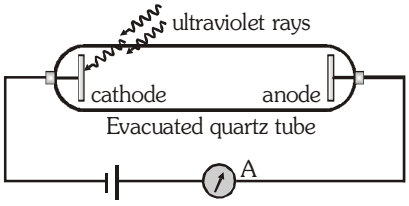
EXPLANATION OF PHOTOELECTRIC EFFECT

- **On the basis of wave theory** :According to wave theory, light is an electromagnetic radiation consisting of oscillating electric field vectors and magnetic field vectors. When electromagnetic radiations are incident on a metal surface, the free electrons [free electrons means the electrons which are loosely bound and free to move inside the metal] absorb energy from the radiation. This occurs by the oscillations of electron under the action of electric field vector of electromagnetic radiation. When an electron acquires sufficiently high energy so that it can overcome its binding energy, it comes out from the metal.
- **On the basis of photon theory:** According to photon theory of light, light consists of particles (called photons). Each particle carries a certain amount of energy with it. This energy is given by $E=h\nu$, where h is the Plank's constant and ν is the frequency. When the photons are incident on a metal surface, they collide with electrons. In some of the collisions, a photon is absorbed by an electron. Thus an electron gets energy $h\nu$. If this energy is greater than the binding energy of the electron, it comes out of the metal surface. The extra energy given to the electron becomes its kinetic energy.

EXPERIMENTS

- **Hertz Experiment**

Hertz observed that when ultraviolet rays are incident on negative plate of electric discharge tube then conduction takes place easily in the tube.

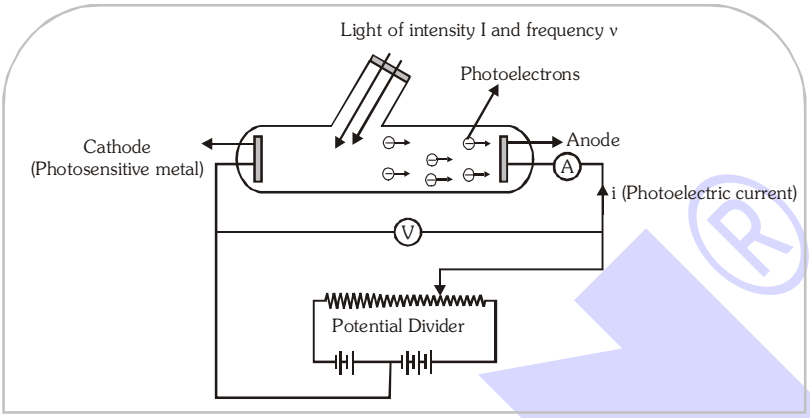


- **Hallwach experiment :** Hallwach observed that if negatively charged Zn plate is illuminated by U.V. light, its negative charge decreases and it becomes neutral and after some time it gains positive charge. It means in the effect of light, some negative charged particles are emitted from the metal.
- **Lenard Explanation :** He told that when ultraviolet rays are incident on cathode, electrons are ejected. These electrons are attracted by anode and circuit is completed due to flow of electrons and current flows. When U.V. rays incident on anode, electrons are ejected but current does not flow.

For the photo electric effect the light of short wavelength (or high frequency) is more effective than the light of long wavelength (or low frequency)

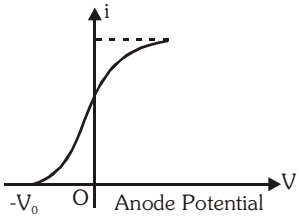
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- **Experimental study of photoelectric Effect :** When light of frequency ν and intensity I falls on the cathode, electrons are emitted from it. The electrons are collected by the anode and a current flows in the circuit. This current is called photoelectric current. This experiment is used to study the variation of photoelectric current with different factors like intensity, frequency and the potential difference between the anode and cathode.



(i) Variation of photoelectric current with potential difference

With the help of the above experimental setup, a graph is obtained between current and potential difference. The potential difference is varied with the help of a potential divider. The graph obtained is shown below.



The main points of observation are :

(a) At zero anode potential, a current exists. It means that electrons are emitted from cathode with some kinetic energy.

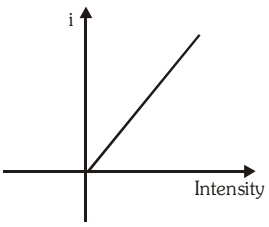
(b) As anode potential is increased, current increases. This implies that different electrons are emitted with different kinetic energies.

(c) After a certain anode potential, current acquires a constant value called saturation current. Current acquires a saturation value because the number of electrons emitted7 per second from the cathode are fixed.

(d) At a certain negative potential, the photoelectric current becomes zero. This is called stopping potential (V_0). Stopping potential is a measure of maximum kinetic energy of the emitted electrons. Let KE_{max} be the maximum kinetic energy of an emitted electron, then $KE_{max} = eV_0$.

(ii) Variation of current with intensity

The photoelectric current is found to be directly proportional to intensity of incident radiation.

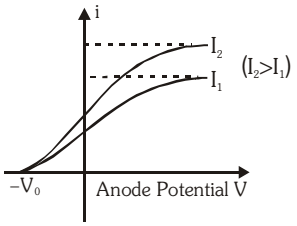


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(iii) Effect of intensity on saturation current and stopping potential

(a) Saturation current increases with increase in intensity.

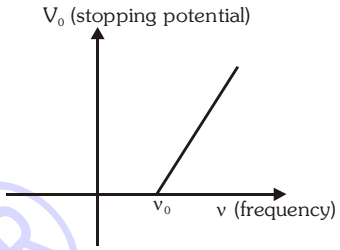
(b) Stopping potential (and therefore maximum kinetic energy) is independent of intensity.



(iv) Effect of frequency

(a) Stopping potential is found to vary with frequency of incident light linearly. Greater the frequency of incident light, greater the stopping potential.

(b) There exists a certain minimum frequency ν_0 below which no stopping potential is required as no emission of electrons takes place. This frequency is called threshold frequency. For photoelectric emission to take place, $\nu > \nu_0$.



IMPORTANT POINTS

- Photo electric effect is an instantaneous process, as soon as light is incident on the metal, photo electrons are emitted.
- Stopping potential does not depend on the distance between cathode and anode.
- The work function represented the energy needed to remove the least tightly bounded electrons from the surface. It depends only on nature of the metal and independent of any other factors.

- **Failure of wave theory of light**

(i) According to wave theory when light incident on a surface, energy is distributed continuously over the surface. So that electron has to wait to gain sufficient energy to come out. But in experiment there is no time lag. Emission of electrons takes place in less than 10^{-9} s. This means, electron does not absorb energy. They get all the energy once.

(ii) When intensity is increased, more energetic electrons should be emitted. So that stopping potential should be intensity dependent. But it is not observed.

(iii) According to wave theory, if intensity is sufficient then, at each frequency, electron emission is possible. It means there should not be existence of threshold frequency.

- **Einstein's Explanation of Photoelectric Effect**

Einstein explained photoelectric effect on the basis of photon–electron interaction. The energy transfer takes place due to collisions between an electrons and a photon. The electrons within the target material are held there by electric force. The electron needs a certain minimum energy to escape from this pull. This minimum energy is the property of target material and it is called the work function. When a photon of energy $E = h\nu$ collides with and transfers its energy to an electron, and this energy is greater than the work function, the electron can escape through the surface.

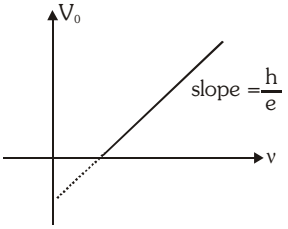
- **Einstein's Photoelectric Equation** $h\nu = \phi + KE_{max}$

Here $h\nu$ is the energy transferred to the electron. Out of this, ϕ is the energy needed to escape.

The remaining energy appears as kinetic energy of the electron.

Now $KE_{max} = eV_0$ (where V_0 is stopping potential)

$$\therefore\ h\nu = \phi + eV_0 \Rightarrow\ V_0 = \left(\frac{h}{e}\right)\nu - \frac{\phi}{e}$$



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Thus, the stopping potential varies linearly with the frequency of incident radiation.

Slope of the graph obtained is $\frac{h}{e}$. This graph helps in determination of Planck's constant.

IMPORTANT POINTS

- Einstein's Photo Electric equation is based on conservation of energy.
- Einstein explained P.E.E. on the basis of quantum theory, for which he was awarded noble prize.
- According to Einstein one photon can eject one e^- only. But here the energy of incident photon should greater or equal to work function.
- In photoelectric effect all photoelectrons do not have same kinetic energy. Their KE range from zero to E_{max} which depends on frequency of incident radiation and nature of cathode.
- The photo electric effect takes place only when photons strike bound electrons because for free electrons energy and momentum conservations do not hold together.

PHOTON THEORY OF LIGHT

- A photon is a particle of light moving with speed 299792458 m/s in vacuum.
- The speed of a photon is independent of frame of reference. This is the basic postulate of theory of relativity.
- The rest mass of a photon is zero. i.e. photons do not exist at rest.
- Effective mass of photon $m = \frac{E}{c^2} = \frac{hc}{c^2\lambda} = \frac{h}{c\lambda}$ i.e. $m \propto \frac{1}{\lambda}$

  So mass of violet light photon is greater than the mass of red light photon. ($\because \lambda_R > \lambda_V$)
- According to Planck the energy of a photon is directly proportional to the frequency of the radiation.

  $E \propto \nu$ or $E = h\nu$

  $E = \frac{hc}{\lambda}$ joule $(\because c = \nu\lambda)$ or $E = \frac{hc}{\lambda e} = \frac{12400}{\lambda}$ eV – Å

  $\left[\because \frac{hc}{e} = 12400(\text{Å} - eV)\right]$

  Here E = energy of photon, c = speed of light, h = Planck's constant, e = charge of electron
  $h = 6.62 \times 10^{-34}$ J–s, ν = frequency of photon, λ = wavelength of photon
- Linear momentum of photon $p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$
- A photon can collide with material particles like electron. During these collisions, the total energy and total momentum remain constant.
- Energy of light passing through per unit area per unit time is known as intensity of light.

  Intensity of light $I = \frac{E}{At} = \frac{P}{A}$ ...(i)

  Here P = power of source, A = Area, t = time taken
  E = energy incident in t time = Nhν, N = number of photon incident in t time

  Intensity $I = \frac{N(h\nu)}{At} = \frac{n(h\nu)}{A}$ ...(ii)

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$\left[\because \quad n = \frac{N}{t} = \text{no. of photon per sec.}\right]$

From equation (i) and (ii), $\frac{P}{A} = \frac{n(h\nu)}{A} \Rightarrow n = \frac{P}{h\nu} = \frac{P\lambda}{hc} = 5 \times 10^{24}\, J^{-1} m^{-1} \times P \times \lambda$

- When photons fall on a surface, they exert a force and pressure on the surface. This pressure is called radiation pressure.

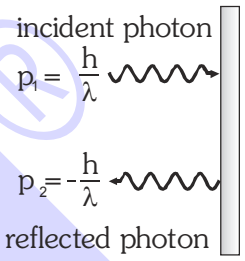
- **Force exerted on perfectly reflecting surface**

Let 'N' photons are there in time t,

Momentum before striking the surface $(p_1) = \frac{Nh}{\lambda}$

Momentum after striking the surface $(p_2) = -\frac{Nh}{\lambda}$

Change in momentum of photons $= p_2 - p_1 = \frac{-2Nh}{\lambda}$



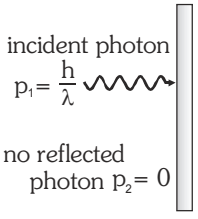
But change in momentum of surface = $\Delta p = \frac{2Nh}{\lambda}$

So that force on surface $F = \frac{2Nh}{t\lambda} = n\left[\frac{2h}{\lambda}\right]$ but $n = \frac{P\lambda}{hc}$

$\therefore F = \frac{2h}{\lambda} \times \frac{P\lambda}{hc} = \frac{2P}{c}$ and Pressure $= \frac{F}{A} = \frac{2P}{cA} = \frac{2I}{c}\left[\because I = \frac{P}{A}\right]$

- **Force exerted on perfectly absorbing surface**

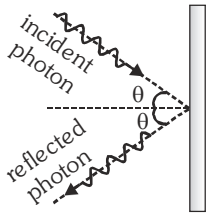
$F = \frac{p_1 - p_2}{t} = \frac{\frac{Nh}{\lambda} - 0}{t} = \frac{Nh}{t\lambda} = n\frac{h}{\lambda}$



$F = \frac{P}{c}\left(\because n = \frac{P\lambda}{hc}\right)$ and Pressure $= \frac{F}{A} = \frac{P}{Ac} = \frac{I}{c}$

- **When a beam of light is incident at angle θ on perfectly reflector surface**

$F = \frac{2P}{c}\cos\theta = n\left[\frac{2h}{\lambda}\right]\cos\theta = \frac{2IA\cos\theta}{c}$



Pressure $= \frac{F}{A\sec\theta} = \frac{2I\cos^2\theta}{c}$

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MATTER WAVES THEORY

DUAL NATURE OF LIGHT

Experimental phenomena of light reflection, refraction, interference, diffraction are explained only on the basis of wave theory of light. These phenomena verify the wave nature of light. Experimental phenomena of light photoelectric effect and Crompton effect, pair production and positron inhalational can be explained only on the basis of the particle nature of light. These phenomena verify the particle nature of light.

It is inferred that light does not have any definite nature, rather its nature depends on its experimental phenomenon. This is known as the dual nature of light. The wave nature and particle nature both can not be possible simultaneously.

De-Broglie HYPOTHESIS

De Broglie imagined that as light possess both wave and particle nature, similarly matter must also posses both nature, particle as well as wave. De Broglie imagined that despite particle nature of matter, waves must also be associated with material particles. Wave associated with material particles, are defined as matter waves.

- **De Broglie wavelength associated with moving particles**

If a particle of mass m moving with velocity v

Kinetic energy of the particle $E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$ momentum of particle $p = mv = \sqrt{2mE}$ the wave length associated with the particles is $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$ $\lambda \propto \frac{1}{p} \Rightarrow \lambda \propto \frac{1}{v} \Rightarrow \lambda \propto \frac{1}{\sqrt{E}}$

The order of magnitude of wave lengths associated with macroscopic particles is 10^{-24} Å.

The smallest wavelength whose measurement is possible is that of γ – rays ($\lambda \simeq 10^{-5}$ Å). This is the reason why the wave nature of macroscopic particles is not observable.

The wavelength of matter waves associated with the microscopic particles like electron, proton, neutron, α – particle, atom, molecule etc. is of the order of 10^{-10} m, it is equal to the wavelength of X–rays, which is within the limit of measurement. Hence the wave nature of these particles is observable.

- **De Broglie wavelength associated with the charged particles**

Let a charged particle having charge q is accelerated by potential difference V.

Kinetic energy of this particle $E = \frac{1}{2}mv^2 = qV$ Momentum of particle $p = mv = \sqrt{2mE} = \sqrt{2mqV}$

The De Broglie wavelength associated with charged particle $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$

- **For an Electron** $m_e = 9.1 \times 10^{-31}$ kg, $q = 1.6 \times 10^{-19}$ C, $h = 6.62 \times 10^{-34}$ J–s

De Broglie wavelength associated with electron $\lambda = \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} V}}$

$\lambda = \frac{12.27 \times 10^{-10}}{\sqrt{V}}$ meter $= \frac{12.27}{\sqrt{V}}$ Å so $\lambda \propto \frac{1}{\sqrt{V}}$

Potential difference required to stop an electron of wavelength λ is $V = \frac{150.6}{\lambda^2}$ volt $(Å)^2$

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- **For Proton** $m_p = 1.67 \times 10^{-27}$ kg
  De Broglie wavelength associated with proton

$$\lambda_p = \frac{6.62\times10^{-34}}{\sqrt{2\times1.67\times10^{-27}\times1.6\times10^{-19}V}} ; \lambda_p = \frac{0.286\times10^{-10}}{\sqrt{V}} \text{ meter} = \frac{0.286}{\sqrt{V}} \text{ Å}$$

- **For Deuteron** $m_d = 2 \times 1.67 \times 10^{-27}$ kg, $q_d = 1.6 \times 10^{-19}$ C

$$\lambda_d = \frac{6.62\times10^{-34}}{\sqrt{2\times2\times1.67\times10^{-27}\times1.6\times10^{-19}V}} = \frac{0.202}{\sqrt{V}} \text{ Å}$$

- **For α Particles** $q_\alpha = 2 \times 1.6 \times 10^{-19}$ C, $m_\alpha = 4 \times 1.67 \times 10^{-27}$ kg

$$\therefore \lambda_\alpha = \frac{6.62\times10^{-34}}{\sqrt{2\times4\times1.67\times10^{-27}\times2\times1.6\times10^{-19}V}} = \frac{0.101}{\sqrt{V}} \text{ Å}$$

DE BROGLIE WAVELENGTH ASSOCIATED WITH UNCHARGED PARTICLES

- **Kinetic energy of uncharged particle** $E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$

  m = mass of particle, v = velocity of particle, p = momentum of particle.

- **Velocity of uncharged particle** $v = \sqrt{\frac{2E}{m}}$

- **Momentum of particle** $p = mv = \sqrt{2mE}$

  wavelength associated with the particle $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$

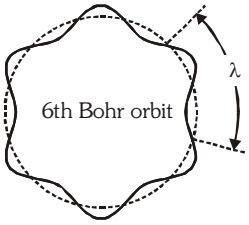
  Kinetic energy of the particle in terms of its wavelength $E = \frac{h^2}{2m\lambda^2} = \frac{h^2}{2m\lambda^2\times1.6\times10^{-19}} eV$

  For a neutron $m_n = 1.67 \times 10^{-27}$ kg

$$\therefore \lambda = \frac{6.62\times10^{-34}}{\sqrt{2\times1.67\times10^{-27}\times E}} = \frac{0.286\times10^{-10}}{\sqrt{E}} \text{meter}\sqrt{eV} = \frac{0.286}{\sqrt{E}} \text{Å}\sqrt{eV}$$

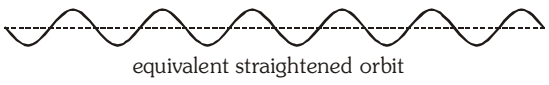
EXPLANATION OF BOHR QUANTIZATION CONDITION

According to De Broglie electron revolves round the nucleus in the form of stationary waves (i. e. wave packet) in the similar fashion as stationary waves in a vibrating string. Electron can stay in those circular orbits whose circumference is an integral multiple of De–Broglie wavelength associated with the electron, $2\pi r = n\lambda$



$\because \lambda = \frac{h}{mv}$ and $2\pi r = n\lambda \therefore mvr = \frac{nh}{2\pi}$

This is the Bohr quantizations condition.



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ATOMIC STRUCTURE

VARIOUS MODELS FOR STRUCTURE OF ATOM

- **Dalton's Theory**

  Every material is composed of minute particles known as atom. Atom is indivisible i.e. it cannot be subdivided. It can neither be created nor be destroyed.

  All atoms of same element are identical physically as well as chemically, whereas atoms of different elements are different in properties.

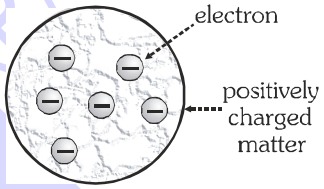
  The atoms of different elements are made up of hydrogen atoms. (The radius of the heaviest atom is about 10 times that of hydrogen atom and its mass is about 250 times that of hydrogen).

  The atom is stable and electrically neutral.

- **Thomson's Atom Model**

  The atom as a whole is electrically neutral because the positive charge present on the atom (sphere) is equal to the negative charge of electrons present in the sphere.

  Atom is a positively charged sphere of radius 10^{-10} m in which electrons are embedded in between.

  The positive charge and the whole mass of the atom is uniformly distributed throughout the sphere.

- **Shortcomings of Thomson's model**

  (i) The spectrum of atoms cannot be explained with the help of this model

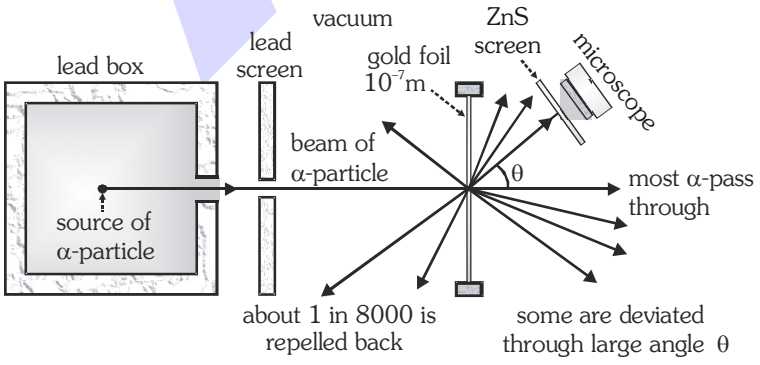
  (ii) Scattering of α–particles cannot be explained with the help of this model

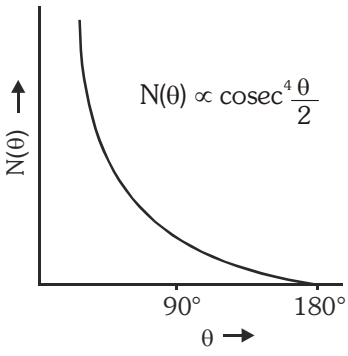
  (iii) Angular frequency of electron in n^{th} orbit $\omega_n = \frac{8\pi^2k^2Z^2e^4m}{n^3h^3} \Rightarrow \omega_n \propto \frac{Z^2m}{n^3}$

RUTHERFORD ATOM MODEL

- **Rutherford experiments on scattering of α – particles by thin gold foil**

  The experimental arrangement is shown in figure. α–particles are emitted by some radioactive material (polonium), kept inside a thick lead box. A very fine beam of α–particles passes through a small hole in the lead screen. This well collimated beam is then allowed to fall on a thin gold foil. While passing through the gold foil, α–particles are scattered through different angles. A zinc sulphide screen was placed out the other side of the gold foil. This screen was movable, so as to receive the α–particles, scattered from the gold foil at angles varying from 0 to 180°. When an α–particle strikes the screen, it produces a flash of light and it is observed by the microscope. It was found that :





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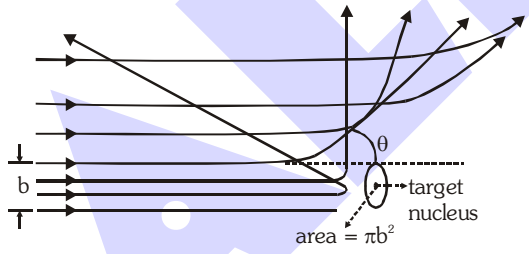
- Most of the α – particles went straight through the gold foil and produced flashes on the screen as if there were nothing inside gold foil. Thus the atom is hollow.
- Few particles collided with the atoms of the foil which have scattered or deflected through considerable large angles. Few particles even turned back towards source itself.
- The entire positive charge and almost whole mass of the atom is concentrated in small centre called a nucleus.
- The electrons could not deflected the path of a α – particles i.e. electrons are very light.
- Electrons revolve round the nucleus in circular orbits. So, Rutherford 1911, proposed a new type of model of the atom. According to this model, the positive charge of the atom, instead of being uniformly distributed throughout a sphere of atomic dimension is concentrated in a very small volume (Less than 10^{-13}m is diameter) at it centre. This central core, now called nucleus, is surrounded by clouds of electron makes.

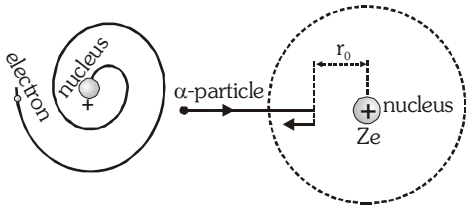
  The entire atom electrically neutral. According to Rutherford scattering formula, the number of α – particle scattered at an angle θ by a target are given by

$$N_\theta = \frac{N_0 nt(2Ze^2)^2}{16(4\pi\varepsilon_0)^2 r^2 (mv_0^2)^2} \times \frac{1}{\sin^4 \frac{\theta}{2}}$$

Where N_0 = number of α – particles that strike the unit area of the scatter

n	=	Number of target atom per m^3
t	=	Thickness of target
Ze	=	Charge on the target nucleus
2e	=	Charge on α – particle
r	=	Distance of the screen from target
v_0	=	Velocity of α – particles at nearer distance of approach the size of a nucleus or the distance of nearer approach is given by.





$$r_0 = \frac{1}{4\pi\varepsilon_0} \times \frac{(2Ze)^2}{\left[\frac{1}{2}mv_0^2\right]} = \frac{1}{4\pi\epsilon_0}\frac{(2Ze)^2}{E_K}$$ where E_K = K.E. of α–particle

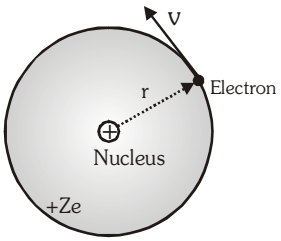
Bohr's Atomic Model

In 1913 Neils Bohr, a Danish Physicist, introduced a revolutionary concept i.e., the quantum concept to explain the stability of an atom. He made a simple but bold statement that "The old classical laws which are applicable to bigger bodies cannot be directly applied to the sub–atomic particles such as electrons or protons.

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Bohr incorporated the following new ideas now regarded as postulates of Bohr's theory.

1. The centripetal force required for an encircling electron is provided by the electrostatic attraction between the nucleus and the electron i.e. $\frac{1}{4\pi\varepsilon_0}\frac{(Ze)e}{r^2}=\frac{mv^2}{r}$ ...(i)



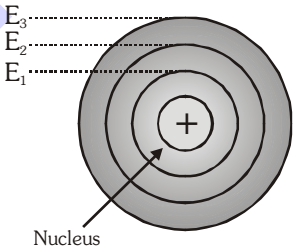
ε_0 =Absolute permittivity of free space = 8.85×10^{-12} C^2 N^{-1} m^{-2}
m = Mass of electron
v = Velocity (linear) of electron
r = Radius of the orbit in which electron is revolving.
Z = Atomic number of hydrogen like atom.

2. Electrons can revolve only in those orbits in which angular momentum of electron about nucleus is an integral multiple of $\frac{h}{2\pi}$. i.e., $mvr=\frac{nh}{2\pi}$ ...(ii)

n = Principal quantum number of the orbit in which electron is revolving.

3. Electrons in an atom can revolve only in discrete circular orbits called stationary energy levels (shells). An electron in a shell is characterised by a definite energy, angular momentum and orbit number. While in any of these orbits, an electron does not radiate energy although it is accelerated.

4. Electrons in outer orbits have greater energy than those in inner orbits. The orbiting electron emits energy when it jumps from an outer orbit (higher energy states) to an inner orbit (lower energy states) and also absorbs energy when it jumps from an inner orbit to an outer orbit. $E_n - E_m = h\nu_{n,m}$



where, E_n = Outer energy state
E_m = Inner energy state
$\nu_{n,m}$ = Frequency of radiation

5. The energy absorbed or released is always in the form of electromagnetic radiations.

MATHEMATICAL ANALYSIS OF BOHR'S THEORY

From above equation (i) and (ii) i.e., $\frac{1}{4\pi\varepsilon_0}\frac{(Ze)e}{r^2}=\frac{mv^2}{r}$ and $mvr=\frac{nh}{2\pi}$ ...(ii)

We get the following results.

1. **Velocity of electron in nth orbit :** By putting the value of mvr in equation (i) from (ii) we get

$$\frac{1}{4\pi\varepsilon_0}Ze^2=\left(\frac{nh}{2\pi}\right)\times v \Rightarrow v=\frac{Z}{n}\left[\frac{e^2}{2\varepsilon_0 h}\right]=\frac{Z}{n}v_0 \text{ ....(iii)}$$

$$\text{Where, } v_0=\frac{\left(1.6\times10^{-19}\right)^2}{2\times8.85\times10^{-12}\times6.625\times10^{-34}}=2.189\times10^6 \text{ ms}^{-1}=\frac{c}{137}=2.2\times10^6 \text{ m/s}$$

where $c = 3\times10^8$ m/s = speed of light in vacuum

2. **Radius of the nth orbit :**

From equation (iii), putting the value of v in equation (ii), we get

$$m\left(\frac{Z}{n}\times\frac{e^2}{2\varepsilon_0 h}\right)r=\frac{nh}{2\pi}\Rightarrow r=\frac{n^2}{Z}\left[\frac{\varepsilon_0 h^2}{\pi me^2}\right]=\frac{n^2}{Z}r_0 \text{ ...(iv)}$$

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where $r_0 = \frac{8.85\times10^{-12}\times\left(6.625\times10^{-34}\right)^2}{3.14\times9.11\times10^{-31}\times\left(1.6\times10^{-19}\right)^2} = 0.529\times10^{-10}$ m ≈ 0.53Å

3. Total energy of electron in nth orbit :

From equation (i) K.E. $= \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\varepsilon_0 r}$ and PE $= \frac{1}{4\pi\varepsilon_0}\frac{(Ze)(-e)}{r} = -2$K.E. $\therefore$ |PE| = 2 KE

Total energy of the system E = KE + PE = –2KE + KE = – KE $= \frac{-Ze^2}{8\pi\varepsilon_0 r}$

By putting the value of r from the equation (iv), we get $E = \frac{Z^2}{n^2}\left(-\frac{me^4}{8\varepsilon_0^2h^2}\right) = \frac{Z^2}{n^2}\cdot E_0$ ...(v)

where $E_0 = \frac{-\left(9.11\times10^{-3}\right)\left(1.6\times10^{-19}\right)^4}{8\times\left(8.85\times10^{-12}\right)^2\times\left(6.625\times10^{-34}\right)^2} = -13.6$ eV

4. Time period of revolution of electron in nth orbit : $T = \frac{2\pi r}{v}$

By putting the values of r and v, from (iii) and (iv) $T = \frac{n^3}{Z^2}\times\left(\frac{4\in_0^2 h^3}{me^4}\right) = \frac{n^3}{Z^2}\cdot T_0$

where, $T_0 = \frac{4\times\left(8.85\times10^{-12}\right)^2\times\left(6.625\times10^{-34}\right)^3}{9.11\times10^{-31}\times\left(1.6\times10^{-19}\right)^4} = 1.51\times10^{-16}$ second

5. Frequency of revolution in nth orbit :

$f = \frac{1}{T} = \frac{Z^2}{n^3}\times\frac{me^4}{4\varepsilon_0^2h^3} = \frac{Z^2}{n^3}\cdot f_0$ where, $f_0 = \frac{9.11\times10^{-31}\times\left(1.6\times10^{-19}\right)^4}{4\times\left(8.85\times10^{-12}\right)^2\left(6.625\times10^{-34}\right)^3} = 6.6\times10^{15}$ Hz

6. Wavelength of photon

$$\Delta E = E_{n_2} - E_{n_1} = \frac{me^4}{8\varepsilon_0^2h^2}\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]Z^2 = 13.6\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]Z^2 \Rightarrow \Delta E = \frac{hc}{\lambda} \Rightarrow \frac{1}{\lambda} = \bar{\nu} = \frac{me^4}{8\varepsilon_0^2h^3c}\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]Z^2$$

$= R_\infty\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]Z^2$ where, $\bar{\nu}$ is called wave number. $R_\infty = R_H$ = Rydberg constant

$= \frac{9.11\times10^{-31}\times\left(1.6\times10^{-19}\right)^4}{8\times\left(8.85\times10^{-12}\right)^2\left(6.625\times10^{-34}\right)^3\times3\times10^8} = 1.097\times10^7\ m^{-1} = 1.097\times10^{-3}\ Å^{-1}$ (for stationary nucleus).

If nucleus is not stationary (i.e., mass of nucleus is not much greater than the mass of the revolving particle like electron), then $R = \frac{R_\infty}{1+m/M}$ where, m = mass of revolving particle and M = mass of nucleus

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SPECTRAL SERIES OF HYDROGEN ATOM

It has been shown that the energy of the outer orbit is greater than the energy of the inner ones. When the Hydrogen atom is subjected to external energy, the electron jumps from lower energy state i.e. the hydrogen atom is excited. The excited state is unstable hence the electron return to its ground state in about 10^{-8} sec. The excess of energy is now radiated in the form of radiations of different wavelength. The different wavelength constitute spectral series. Which are characteristic of atom emitting, then the wave length of different members of series can be found from the following relations

$$\bar{\nu} = \frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

This relation explains the complete spectrum of hydrogen. A detailed account of the important radiations are listed below.

- **Lyman Series** : The series consist of wavelength which are emitted when electron jumps from an outer orbits to the first orbit i. e., the electronic jumps to K orbit give rise to lyman series. Here $n_1 = 1$ & $n_2 = 2, 3, 4, ...\infty$

The wavelengths of different members of Lyman series are :

- **First member** : In this case $n_1 = 1$ and $n_2 = 2$

$$\text{hence } \frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = \frac{3R}{4} \Rightarrow \lambda = \frac{4}{3R} \Rightarrow \lambda = \frac{4}{3\times 10.97\times 10^6} = 1216 \times 10^{-10}\ \text{m} = 1216\text{Å}$$

- **Second member :** In this case $n_1 = 1$ and $n_2 = 3$ hence

$$\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{3^2}\right] = \frac{8R}{9} \Rightarrow \lambda = \frac{9}{8R} \Rightarrow \lambda = \frac{9}{8\times 10.97\times 10^6} = 1026 \times 10^{-10}\ \text{m} = 1026\text{Å}$$

Similarly the wavelength of the other members can be calculated.

- **Limiting members** : In this case $n_1 = 1$ and $n_2 = \infty$, hence

$$\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right] = R \Rightarrow \lambda = \frac{1}{R} \Rightarrow \lambda = \frac{1}{10.97\times 10^6} = 912 \times 10^{-10}\text{m} = 912\text{Å}$$

This series lies in ultraviolet region.

- **Balmer Series :** This series is consist of all wave lengths which are emitted when an electron jumps from an outer orbit to the second orbit i. e., the electron jumps to L orbit give rise to Balmer series. Here $n_1 = 2$ and $n_2 = 3, 4, 5$ .....∞ The wavelength of different members of Balmer series.

- **First member** : In this case $n_1 = 2$ and $n_2 = 3$, hence

$$\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = \frac{5R}{36} \Rightarrow \lambda = \frac{36}{5R} \Rightarrow \lambda = \frac{36}{5\times 10.97\times 10^6} = 6563 \times 10^{-10}\text{m} = 6563\text{Å}$$

- **Second member** : In this case $n_1 = 2$ and $n_2 = 4$, hence.

$$\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = \frac{3R}{16} \Rightarrow \lambda = \frac{16}{3R} \Rightarrow \lambda = \frac{16}{3\times 10.97\times 10^6} = 4861 \times 10^{-10}\text{m} = 4861\text{Å}$$

- **Limiting members**: In this case $n_1 = 2$ and $n_2 = \infty$, hence $\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{\infty}\right] = \frac{R}{4} \Rightarrow \lambda = \frac{4}{R} = 3646\text{Å}$

This series lies in visible and near ultravoilet region.

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- **Paschen Series :** This series consist of all wavelength are emitted when an electron jumps from an outer orbit to the third orbit i. e., the electron jumps to M orbit give rise to paschen series. Here n_1=3 & n_2= 4, 5, 6 .....∞. The different wavelengths of this series can be obtained from the formula

$$\frac{1}{\lambda} = R\left[\frac{1}{3^2} - \frac{1}{n_2^2}\right]$$

where n_2 = 4, 5, 6 .....∞
For the first member, the wavelengths is 18750Å. This series lies in infra–red region.

- **Brackett Series :** This series is consist of all wavelengths which are emitted when an electron jumps from an outer orbits to the fourth orbit i. e., the electron jumps to N orbit give rise to Brackett series. Here n_1 = 4 & n_2 = 5, 6, 7, ...∞.

The different wavelengths of this series can be obtained from the formula $\frac{1}{\lambda} = R\left[\frac{1}{4^2} - \frac{1}{n_2^2}\right]$

where n_2 = 5, 6, 7 .....∞
This series lies in infra–red region of spectrum.

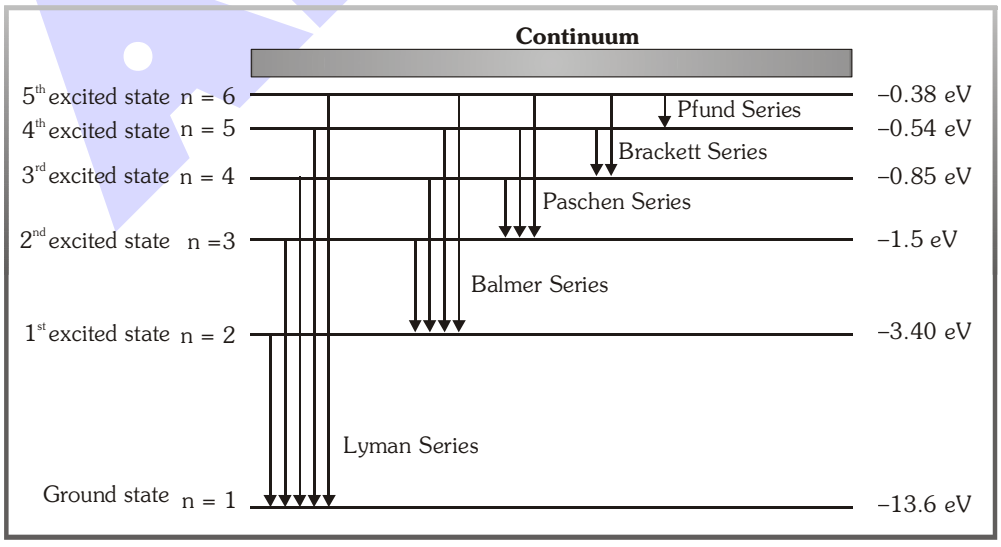
- **Pfund series :** The series consist of all wavelengths which are emitted when an electron jumps from an outer orbit to the fifth orbit i. e., the electron jumps to O orbit give right to Pfund series. Here n_1 = 5 and n_2 = 6, 7, 8 .....∞.

The different wavelengths of this series can be obtained from the formula $\frac{1}{\lambda} = R\left[\frac{1}{5^2} - \frac{1}{n_2^2}\right]$

where n_2 = 6, 7, 8 .....∞
This series lies in infra–red region of spectrum.

The result are tabulated below

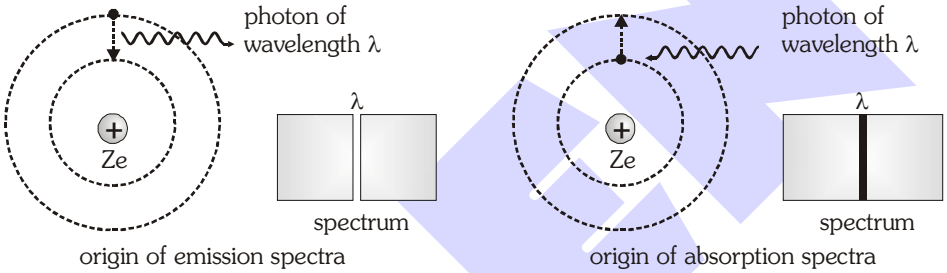
S. No.	Series Observed	Value of n_1	Value of n_2	Position in the Spectrum
1.	Lyman Series	1	2,3,4...∞	Ultra Voilet
2.	Balmer Series	2	3,4,5...∞	Visible
3.	Paschen Series	3	4,5,6....∞	Infra–red
4.	Brackett Series	4	5,6,7....∞	Infra–red
5.	Pfund Series	5	6,7,8....∞	Infra–red



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EXCITATION AND IONISATION OF ATOMS

Consider the case of simplest atom i. e., hydrogen atom, this has one electron in the innermost orbit i.e., (n = 1) and is said to be in the unexcited or normal state. If by some means, sufficient energy is supplied to the electron. It moves to higher energy states. When the atom is in a state of a high energy it is said to be excited. The process of raising or transferring the electron from lower energy state is called excitation. When by the process of excitation, the electron is completely removed from the atom. The atom is said to be ionized. Now the atom has left with a positive charge. Thus the process of raising the atom from the normal state to the ionized state is called ionisation. The process of excitation and ionisation both are absorption phenomena. The excited state is not stationary state and lasts in a very short interval of time (10^{-8} sec) because the electron under the attractive force of the nucleus jumps to the lower permitted orbit. This is accompanied by the emission of radiation according to BOHR'S frequency condition.



The energy necessary to excite an atom can be supplied in a number of ways. The most commonly kinetic energy (Wholly or partly) of the electrons is transferred to the atom. The atom is now in a excited state. The minimum potential V required to accelerate the bombarding electrons to cause excitation from the ground state is called the resonance potential. The various values of potential to cause excitation of higher state called **excitation potential**. The potential necessary to accelerate the bombarding electrons to cause ionisation is called the **ionization potential**. The term critical potential is used to include the resonance, excitation and ionisation potential. We have seen that the energy required to excite the electron from first to second state is 13.6 – 3.4 = 10.2 eV. from first to third state is 13.6 – 1.5 = 12.1 eV., and so on. The energy required to ionise hydrogen atom is 0 – (– 13.6) = 13.6 eV. Hence ionization potential of hydrogen atom is 13.6 volt.

SUCCESSES AND LIMITATIONS

Bohr showed that Planck's quantum ideas were a necessary element of the atomic theory. He introduced the idea of quantized energy levels and explained the emission or absorption of radiations as being due to the transition of an electron from one level to another. As a model for even multielectron atoms, the Bohr picture is still useful. It leads to a good, simple, rational ordering of the electrons in larger atoms and quantitatively helps to predict a greater deal about chemical behavior and spectral detail.

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Bohr's theory is unable to explain the following facts :

- The spectral lines of hydrogen atom are not single lines but each one is a collection of several closely spaced lines.
- The structure of multielectron atoms is not explained.
- No explanation for using the principles of quantization of angular momentum.
- No explanation for Zeeman effect. If a substance which gives a line emission spectrum is placed in a magnetic field, the lines of the spectrum get splitted up into a number of closely spaced lines. This phenomenon is known as Zeeman effect.

Ex. A hydrogen like atom of atomic number Z is in an excited state of quantum number 2n. It can emit a maximum energy photon of 204 eV. If it makes a transition to quantum state n, a photon of energy 40.8 eV is emitted. Find n, Z and the ground state energy (in eV) for this atom. Also, calculate the minimum energy (in eV) that can be emitted by this atom during de–excitation. Ground state energy of hydrogen atom is –13.6 eV.

Sol. The energy released during de–excitation in hydrogen like atoms is given by :

$$E_{n_2} - E_{n_1} = \frac{me^4}{8\varepsilon_0^2 h^2}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]Z^2$$

Energy released in de–excitation will be maximum if transition takes place from nth energy level to ground state i.e.,

$$E_{2n} - E_1 = 13.6\left[\frac{1}{1^2} - \frac{1}{(2n)^2}\right]Z^2 = 204 \text{ eV} \quad ...(i) \quad \& \quad \text{also } E_{2n} - E_n = 13.6\left[\frac{1}{n^2} - \frac{1}{(2n)^2}\right]Z^2 = 40.8 \text{ eV} \quad ...(ii)$$

Taking ratio of (i) to (ii), we will get $\frac{4n^2 - 1}{3} = 5 \Rightarrow n^2 = 4 \Rightarrow n = 2$

Putting n=2 in equation (i) we get

$$Z^2 = \frac{204 \times 16}{13.6 \times 15} \Rightarrow Z = 4 \quad \because E_n = -13.6\,\frac{Z^2}{n^2} \Rightarrow E_1 = -13.6 \times \frac{4^2}{1^2} = -217.6 \text{ eV} = \text{ground state energy}$$

ΔE is minimum if transition will be from 2n to 2n–1 i.e. between last two adjacent energy levels.

$$\therefore \Delta E_{min} = E_{2n} - E_{2n-1} = 13.6\left[\frac{1}{3^2} - \frac{1}{4^2}\right]4^2 = 10.57 \text{ eV}$$

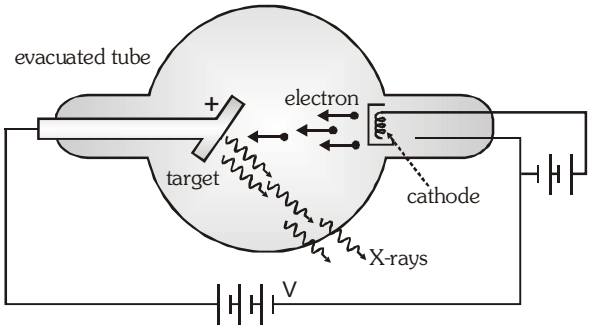
is the minimum amount of energy released during de–excitation.

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X–RAYS

ROENTGEN EXPERIMENT

Roentgen discovered X–ray. While performing experiment on electric discharge tube Roentgen observed that when pressure inside the tube is 10^{-3}mm of Hg and applied potential is kept 25kV then some unknown radiation are emitted by anode. These are known as X–ray. X–rays are produced by bombarding high speed electrons on a target of high atomic weight and high melting point.



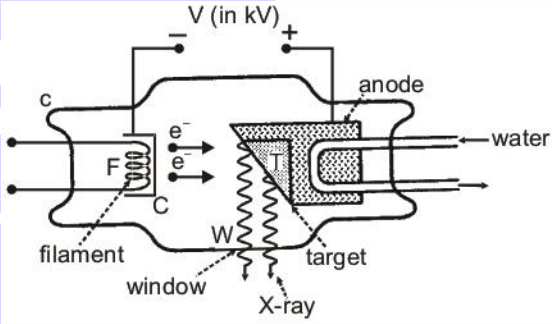
To Produce X–ray Three Things are Required

(i) Source of electron (ii) Means of accelerating these electron to high speed

(iii) Target on which these high speed electron strike

COOLIDGE METHOD

Coolidge developed thermoionic vacuum X–ray tube in which electron are produced by thermoionic emission method. Due to high potential difference electrons (emitted due to thermoionic method) move towards the target and strike from the atom of target due to which X–ray are produced. Experimentally it is observed that only 1% or 2% kinetic energy of electron beam is used to produce X–ray. Rest of energy is wasted in form of heat.



Characteristics of target

(a) Must have high atomic number to produce hard X–rays.

(b) High melting point to withstand high temperature produced.

(c) High thermal conductivity to remove the heat produced

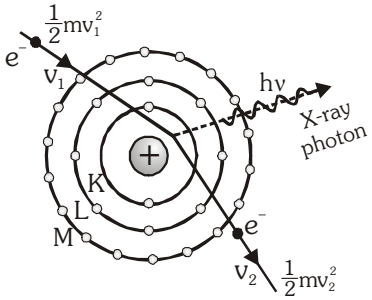
(d) Tantalum, platinum, molybdenum and tungsten serve as target materials

- **Control of intensity** : The intensity of X–ray depends on number of electrons striking the target and number of electron depend on temperature of filament which can be controlled by filament current. Thus intensity of X–ray depends on current flowing through filament.
- **Control of Penetrating Power**: The Penetrating power of X–ray depends on the energy of incident electron. The energy of electron can be controlled by applied potential difference. Thus penetrating power of X–ray depend on applied potential difference. Thus the intensity of X–ray depends on current flowing through filament while penetrating power depends on applied potential difference

	Soft X–ray	Hard X–ray
Wavelength	10 Å to 100 Å	0.1 Å – 10 Å
Energy	$\frac{12400}{\lambda}$ eV–Å	$\frac{12400}{\lambda}$ eV–Å
Penetrating power	Less	More
Use	Radio photography	Radio therapy

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- **Continuous spectrum of X–ray** : When high speed electron collides from the atom of target and passes close to the nucleus.There is coulomb attractive force due to this electron is deaccelerated i.e. energy is decreased. The loss of energy during deacceleration is emitted in the form of X–rays. X–ray produced in this way are called Braking or Bremstralung radiation and form continuous spectrum. In continuous spectrum of X–ray all the wavelength of X–ray are present but below a minimum value of wavelength there is no X–ray. It is called cut off or threshold or minimum wavelength of X–ray. The minimum wavelength depends on applied potential.



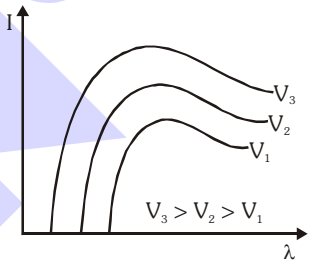
- **Loss in Kinetic Energy**

$\frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 = h\nu$ + heat energy if $v_2 = 0$, $v_1 = v$

(In first collision, heat = 0)

$\frac{1}{2}mv^2 = h\nu_{max}$ ...(i)

$\frac{1}{2}mv^2 = eV$ ...(ii) [here V is applied potential]

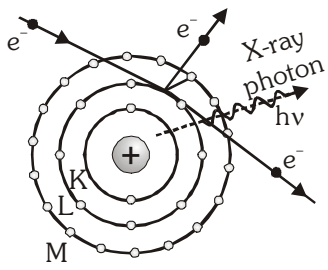


from (i) and (ii) $h\nu_{max} = eV \Rightarrow \frac{hc}{\lambda_{min}} = eV \Rightarrow \lambda_{min} = \frac{12400}{V} \times \text{ volt} = \frac{12400}{V} \times 10^{-10}\text{ m} \times \text{ volt}$

Continuous X–rays also known as white X–ray. Minimum wavelength of these spectrum only depends on applied potential and doesn' t depend on atomic number.

- **Characteristic Spectrum of X–ray**

When the target of X–ray tube is collide by energetic electron it emits two type of X–ray radiation. One of them has a continuous spectrum whose wavelength depend on applied potential while other consists of spectral lines whose wavelength depend on nature of target. The radiation forming the line spectrum is called characteristic X–rays. When highly accelerated electron strikes with the atom of target then it knockout the electron of orbit, due to this a vacancy is created. To fill this vacancy electron jumps from higher energy level and electromagnetic radiation are emitted which form characteristic spectrum of X–ray. Whose wavelength depends on nature of target and not on applied potential.



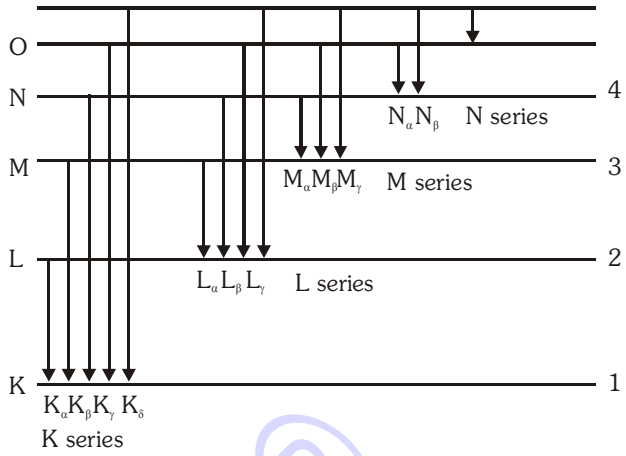
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From Bohr Model

$n_1 = 1$, $n_2 = 2, 3, 4$.....K series

$n_1 = 2$, $n_2 = 3, 4, 5$.....L series

$n_1 = 3$, $n_2 = 4, 5, 6$.....M series

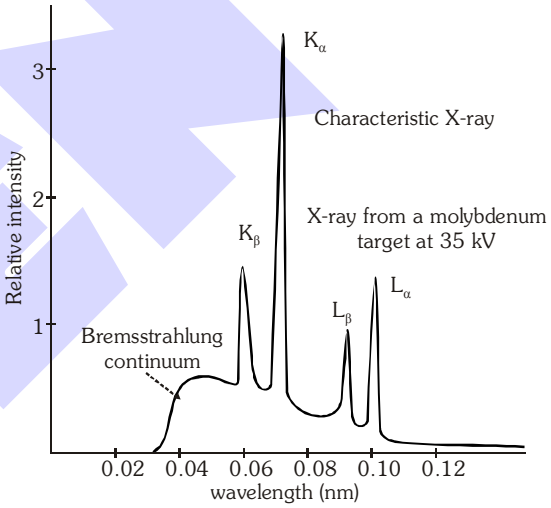


First line of series = α

Second line of series = β

Third line of series = γ

Transition	Wave–length	Energy	Energy difference	Wavelength
$L \to K$	$\lambda_{K\alpha}$	$h\nu_{K\alpha}$	$-(E_K - E_L)$	$\lambda_{K\alpha} = \frac{hc}{(E_K - E_L)}$
$(2 \to 1)$			$= h\nu_{K\alpha}$	$= \frac{12400}{(E_K - E_L)}$ eVÅ
$M \to K$	$\lambda_{K\beta}$	$h\nu_{K\beta}$	$-(E_K - E_M)$	$\lambda_{K\beta} = \frac{hc}{(E_K - E_M)}$
$(3 \to 1)$			$= h\nu_{K\beta}$	$= \frac{12400}{(E_K - E_M)}$ eVÅ
$M \to L$	$\lambda_{L\alpha}$	$h\nu_{L\alpha}$	$-(E_L - E_M)$	$\lambda_{L\alpha} = \frac{hc}{(E_L - E_M)}$
$(3 \to 2)$			$= h\nu_{L\alpha}$	$= \frac{12400}{(E_L - E_M)}$ eVÅ



MOSELEY'S LAW

Moseley studied the characteristic spectrum of number of many elements and observed that the square root of the frequency of a K– line is closely proportional to atomic number of the element. This is

called Moseley's law.

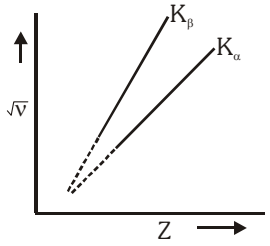
$$\sqrt{\nu} \propto (Z - b) \Rightarrow \nu \propto (Z - b)^2 \Rightarrow \nu = a\,(Z - b)^2 \quad ...(i)$$

Z = atomic number of target

ν = frequency of characteristic spectrum

b = screening constant (for K– series b=1, L series b=7.4)

a = proportionality constant



From Bohr Model $\nu = RcZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$ ...(ii)

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Comparing (i) and (ii) $a = Rc\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$

- Thus proportionality constant 'a' does not depend on the nature of target but depend on transition.

	Bohr model		**Moseley's correction**
1.	For single electron species	**1.**	For many electron species
2.	$\Delta E = 13.6Z^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$ eV	**2.**	$\Delta E = 13.6\ (Z-1)^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$ eV
3.	$\nu = RcZ^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$	**3.**	$\nu = Rc(Z-1)^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$
4.	$\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$	**4.**	$\frac{1}{\lambda} = R\ (Z-1)^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$

- For X–ray production, Moseley formulae are used because heavy metal are used.

When target is same $\lambda \propto \frac{1}{\frac{1}{n_1^2}-\frac{1}{n_2^2}}$ **When transition is same** $\lambda \propto \frac{1}{(Z-b)^2}$

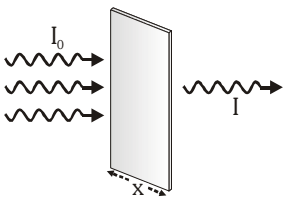
ABSORPTION OF X–RAY

When X–ray passes through x thickness then its intensity $I = I_0e^{-\mu x}$

I_0 = Intensity of incident X–ray

I = Intensity of X–ray after passing through x distance

μ = absorption coefficient of material



- Intensity of X–ray decrease exponentially.
- Maximum absorption of X–ray $\rightarrow$ Lead
- Minimum absorption of X–ray $\rightarrow$ Air

Half thickness ($x_{1/2}$)

The distance travelled by X–ray when intensity become half the original value $x_{1/2} = \frac{\ell n_2}{\mu}$

Ex. When X–rays of wavelength 0.5Å pass through 10 mm thick Al sheet then their intensity is reduced to one sixth. Find the absorption coefficient for Aluminium .

Sol. $\mu = \frac{2.303}{x}\ \log\left(\frac{I_o}{I}\right) = \frac{2.303}{10}\log_{10}6 = \frac{2.303\times 0.7781}{10} = 0.1752/\text{mm}$

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DIFFRACTION OF X–RAY

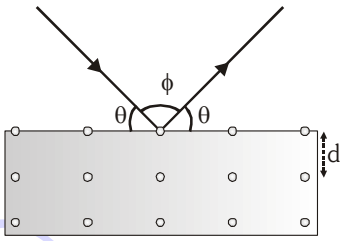
Diffraction of X–ray is possible by crystals because the interatomic spacing in a crystal lattice is order of wavelength of X–rays it was first verified by Lauve.

Diffraction of X–ray take place according to Bragg's law $2d \sin\theta = n\lambda$

d = spacing of crystal plane or lattice constant or distance between adjacent atomic plane

θ = Bragg's angle or glancing angle

ϕ = Diffracting angle n = 1, 2, 3 .....



For Maximum Wavelength

$\sin\theta = 1, n = 1 \Rightarrow \lambda_{max} = 2d$

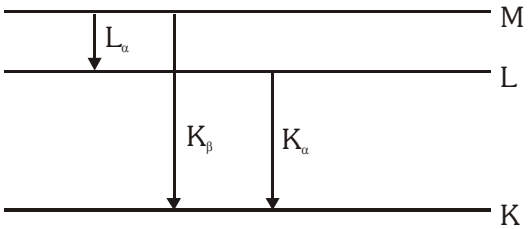
so if $\lambda > 2d$ diffraction is not possible i.e. solution of Bragg's equation is not possible.

PROPERTIES OF X–RAY

- X–ray always travel with the velocity of light in straight line because X–rays are em waves
- X–ray is electromagnetic radiation it show particle and wave both nature
- In reflection, diffraction, interference, refraction X–ray shows wave nature while in photoelectric effect it shows particle nature.
- There is no charge on X–ray thus these are not deflected by electric field and magnetic field.
- X–ray are invisible.
- X–ray affects the photographic plate
- When X–ray incidents on the surface of substance it exerts force and pressure and transfer energy and momentum
- Characteristic X–ray can not obtained from hydrogen because the difference of energy level in hydrogen is very small.

Ex. Show that the frequency of K_β X–ray of a material is equal to the sum of frequencies of K_α and L_α X–rays of the same material.

Sol. The energy level diagram of an atom with one electron knocked out is shown above.



Energy of K_α X–ray is $E_{K\alpha} = E_L - E_K$

and of K_β X–ray is $E_{K\beta} = E_M - E_K$

and of L_α X–rays is $E_{L\alpha} = E_M - E_L$

thus $E_{K\beta} = E_{K\alpha} + E_{L\alpha}$ or $\nu_{K\beta} = \nu_{K\alpha} + \nu_{L\alpha}$

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